



SDN CAMPUS NETWORK NETWORK MONITORING & SECURITY

Official Network Performance and Security Assessment Report

SAVEETHA SCHOOL OF ENGINEERING
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SOFTWARE DEFINED NETWORKING • NETWORK SECURITY • REAL-TIME MONITORING

1. Executive Summary

This report presents the current performance and security condition of the Software Defined Networking (SDN) campus network. The monitoring system continuously observes network throughput, latency, packet loss, active connections and individual link utilization.

The system also performs congestion detection and dynamically selects an alternative network path when the primary link experiences high utilization. This approach supports improved network availability, traffic management and security visibility.

CURRENT NETWORK STATUS

NETWORK OPERATING NORMALLY

2. Network Performance Metrics

METRIC	CURRENT VALUE	STATUS
Throughput	87 Mbps	MONITORED
Latency	45 ms	MONITORED
Packet Loss	3.6 %	MONITORED
Active Connections	125	ACTIVE

The above metrics represent the latest simulated telemetry collected by the SDN campus monitoring system.

3. SDN Network Topology

DEVICE	ROLE	CONNECTED PATH
S1	Core Switch	S1 → S2 / S1 → S3
S2	Access Switch	S2 → S1 / S2 → S3
S3	Access Switch	S3 → S1 / S3 → S2

The network consists of a core switch and access switches with multiple communication paths. The SDN controller can dynamically manage traffic routing according to network conditions.

4. Link Utilization Analysis

NETWORK LINK	UTILIZATION	STATUS
S1 → S2	77 %	MODERATE
S1 → S3	47 %	NORMAL
S2 → S3	54 %	MODERATE

5. QoS & Dynamic Load Balancing

TRAFFIC TYPE	PRIORITY
VoIP	HIGH
Video	HIGH
Web	NORMAL
Download	LOW

The system assigns higher priority to delay-sensitive traffic such as VoIP and video. When congestion is detected on the primary route, the SDN system can select a lower-congestion alternative path.

ROUTING PARAMETER	CURRENT RESULT
Current Network Status	NORMAL
Recommended Traffic Path	S1 → S2
Primary Link Utilization	77 %

6. Cybersecurity Assessment

- **Continuous Monitoring:** Network telemetry should be continuously monitored to identify abnormal traffic conditions.
- **Congestion Detection:** High-utilization links should be identified early to prevent service degradation.
- **Traffic Prioritization:** Critical traffic should receive appropriate QoS priority.
- **Dynamic Routing:** Alternative paths should be used when the primary route becomes congested.
- **Security Visibility:** SDN centralized control can improve network visibility and support security monitoring.

7. Security & Network Recommendations

1. Maintain continuous monitoring of network traffic.
2. Investigate links with utilization above 85%.
3. Use dynamic load balancing during congestion.
4. Maintain QoS priority for critical applications.
5. Monitor packet loss and latency trends.
6. Integrate intrusion detection and prevention mechanisms.
7. Maintain secure communication between SDN components.
8. Regularly review network logs and security events.

8. Conclusion

The SDN Campus Network Monitoring System provides a centralized approach for observing network performance, detecting congestion and dynamically managing traffic. The integration of real-time monitoring, QoS and load balancing improves network visibility and supports security-focused network administration.

The generated assessment demonstrates how SDN technologies can be used to improve campus network reliability, performance and cybersecurity awareness.

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